

What do IO-bound and CPU-bound mean?

- **IO-bound (Input/Output-bound):**

Tasks that spend most of their time **waiting for external operations** like file reads/writes, network requests, database queries, or user input.

The bottleneck is **waiting for I/O operations to complete**, not the CPU itself.

- **CPU-bound:**

Tasks that spend most of their time **doing intensive computations**, using CPU cycles heavily (e.g., heavy math, data processing, image/video encoding).

The bottleneck is the **CPU speed and availability**.

Why is this important in Python?

Python has different strategies to improve performance depending on whether a task is IO-bound or CPU-bound:

- For **IO-bound** tasks, you often use **asynchronous programming** (`asyncio`) or **multithreading** because threads can wait while I/O is happening.
- For **CPU-bound** tasks, **multiprocessing** is usually preferred to bypass Python's Global Interpreter Lock (GIL), which limits true parallel execution in threads.

Examples:

- **IO-bound task:** downloading files, querying a database, reading/writing files, waiting for user input.
- **CPU-bound task:** calculating Fibonacci numbers recursively, image processing, video transcoding, heavy mathematical simulations.

In summary:

Type	Bottleneck	Python tools to speed up
IO-bound	Waiting on I/O	<code>asyncio</code> , threading, non-blocking I/O
CPU-bound	Processor power	<code>multiprocessing</code> , C extensions, PyPy